Diffraction

October 23, 2026

[2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import chi2

#Import data
datafile = pd.read_excel('PHYS 2007 - Diffraction.xlsx',
                                    sheet_name = 'Single Slit Diffraction')
#Display data
display(datafile)
```

	Single Slit difffraction (Dark Spots)	Unnamed: 1	Unnamed: 2	Unnamed: 3
0	Trial 1	Left	NaN	Right
1	NaN	Degrees	Minutes	Degrees
2	Centre	84.5	22	NaN
3	1	85	0	84.5
4	2	85	7	84.5
5	3	85	19	84.5
6	4	85	24	84
7	5	85.5	3	84
8	6	85.5	8	84
9	NaN	NaN	NaN	NaN
10	Trial 2	Left	NaN	Right
11	NaN	Degrees	Minutes	Degrees
12	Centre	84.5	22	NaN
13	1	85	0	84.5
14	2	85	4	84.5
15	3	85	12	84
16	4	85	21	84
17	5	85	30	84
18	6	85.5	5	84
19	NaN	NaN	NaN	NaN
20	Trial 3	Left	NaN	Right
21	NaN	Degrees	Minutes	Degrees
22	Centre	84.5	21	NaN
23	1	85	0	84.5
24	2	85	6	84.5
25	3	85	15	84.5

```
26                                   4          85          21          84
27                                   5        85.5           0          84
28                                   6        85.5          10          84
```

```
   Unnamed: 4
0         NaN
1     Minutes
2         NaN
3          15
4           7
5           0
6          22
7          13
8           5
9         NaN
10        NaN
11    Minutes
12        NaN
13         11
14          1
15         26
16         17
17         10
18          5
19        NaN
20        NaN
21    Minutes
22        NaN
23         12
24          6
25          0
26         21
27         12
28          9
```

[16]:
```
#Grabing numbers code and summing the angle readings
#Single Slit Diffraction (Dark Spots)
#Trial 1 Left
Ddata = datafile['Unnamed: 1'].iloc[3:8].to_numpy()
Mdata = datafile['Unnamed: 2'].iloc[3:8].to_numpy()

phi = np.mean(Ddata + Mdata/60)
#print(phi)

#Trial 1 Right
Ddata2 = datafile['Unnamed: 3'].iloc[3:8].to_numpy()
Mdata2 = datafile['Unnamed: 4'].iloc[3:8].to_numpy()
```

```
phi2 = np.mean(Ddata2 + Mdata2/60)

theta = (Ddata + Mdata/60)-(Ddata2 + Mdata2/60)

#print(phi2)
print(theta)
```

```
[0.25 0.5 0.8166666666666629 1.0333333333333456 1.3333333333333286]
```

[17]:
```
#Grabing numbers code and summing the angle readings
#Single Slit Diffraction (Dark Spots)
#Trial 2 Left
Ddata3 = datafile['Unnamed: 1'].iloc[13:18].to_numpy()
Mdata3 = datafile['Unnamed: 2'].iloc[13:18].to_numpy()

phi3 = np.mean(Ddata3 + Mdata3/60)

#print(phi3)

#Trial 2 Right
Ddata4 = datafile['Unnamed: 3'].iloc[13:18].to_numpy()
Mdata4 = datafile['Unnamed: 4'].iloc[13:18].to_numpy()

phi4 = np.mean(Ddata4 + Mdata4/60)

theta2 = (Ddata3 + Mdata3/60)-(Ddata4 + Mdata4/60)

print(theta2)
```

```
85.22333333333333
[0.3166666666666629 0.5499999999999972 0.7666666666666657
 1.0666666666666629 1.3333333333333286]
```

[14]:
```
#Grabing numbers code and summing the angle readings
#Single Slit Diffraction (Dark Spots)
#Trial 3 Left
Ddata5 = datafile['Unnamed: 1'].iloc[13:18].to_numpy()
Mdata5 = datafile['Unnamed: 2'].iloc[13:18].to_numpy()

phi5 = np.mean(Ddata5 + Mdata5/60)
print(phi5)

#Trial 3 Right
Ddata6 = datafile['Unnamed: 3'].iloc[13:18].to_numpy()
Mdata6 = datafile['Unnamed: 4'].iloc[13:18].to_numpy()

phi6 = np.mean(Ddata6 + Mdata6/60)
```

```
print(phi6)
```

```
85.22333333333333
84.41666666666666
```

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